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1. What is Data ?

Data is a collection of information used for processing, analysis and decision making

Eg :- stud-records, bank transactions customer orders

2. What are the types of data ?

→ structured (Database tables)
→ Semi structured (JSON, XML)
→ Unstructured (Images, videos)

3 What is structured Data ?

structured data is data stored in rows and columns || Eg : Oracle, mysql tables

4 What is semi-structured Data

Semi-structured data does not follow strict table format but has some structure
Eg : JSON, XML

5 What is unstructured data ?

Data without fixed format.
Eg: videos, images, audio, social media posts

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6 What is BigData
Bigdata is extremely large and complex data that traditional systems cannot process efficiently.

Why ?
To solve → storage problems
processing problems
scalability issues

7 What are 3Vs of bigdata

Volume → Huge amount of data

Velocity → Speed of incoming data

Variety → Different data formats

8 What is volume ?
volume means huge amount of data
Eg: youtube videos, banking records

9 What is velocity ?
velocity means speed at which data is generated
eg: live transactions
GPS tracking

10 What is Variety ?
Variety means different forms of data
Eg → tables, videos, JSON, logs.

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11 What is Distributed System

A distributed system uses multiple machines together to store and process data.

Why ?
Bcz, 1 machine cannot efficiently handle
→ huge storage
→ huge processing
→ high traffic

12 What is Hadoop

Hadoop is a framework used for distributed storage and distributed processing of huge data using multiple machines.

→ Main components of Hadoop

HDFS (storage)

MapReduce (processing)

→ Features of Hadoop
Distributed storage
Distributed processing
Scalability
Fault tolerance

13 Why hadoop is scalable

BCz, we can easily add more machines to the cluster

Distributed System → IDEA

Hadoop Framework → Implementation

Tech.. provides for

- Software
- Architecture
- Tools

Many technologies use distributed systems

- Hadoop
- Spark
- Kafka
- Cassandra
- Kubernetes

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What is File System

file system is a method used to store and organize files.

Types

Local file system
Network file system
Distributed file system

Local file system

file accessible only in same machine

Network file system

Multiple machines can access files through network

Distributed file system

Data itself is stored across many machines

It stores files across multiple machines.

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What is Block in Hadoop?

A Block is a small piece of a large file.

Why hadoop splits files into blocks.?

distributed storage
parallel processing
Easier management

What is default block size in HDFS? ⟹ 128 MB

What is HDFS?

HDFS stands for Hadoop Distributed File System

used to store huge files across multiple machines.

Main components of HDFS
NameNode
DataNode

What is NameNode?

NameNode is the master node in HDFS that manages metadata and block information

What is DataNode?

DataNode stores actual data blocks in HDFS.

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What is Metadata ?

Metadata means information about data.

eg:- file name, file size, block locations

What is Replication ?

Replication means creating multiple copies of data blocks for fault tolerance.

It provides → Data safety
→ fault tolerance
(If one machine fails, another copy available).

What is fault tolerance ?

Fault tolerance means system continues working even if some machines fail.

What is HDFS write flow ?

HDFS writes flow is the process of storing a file into HDFS.

Explain HDFS write flow

1. client sends request to NameNode
2. NameNode gives block locations
3. File splits into blocks
4. Client sends blocks to DataNodes
5. Replication happens
6. File stored successfully.

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Note :- client sends actual data directly to DataNode.

What is HDFS Read flow

It is the process of reading a file from HDFS

Explain HDFS Read flow

client requests file from NameNode.
NameNode provides block locations
client Contacts DataNodes
DataNodes send blocks
client combines blocks

Note :-

NameNode only provides block locations not actual data during read flow.

DataNode sends actual data to client

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What is MapReduce

MapReduce is a distributed processing framework used to process huge data in Hadoop.

Phases in MapReduce

→ Map phase
→ Reduce phase

What is Mapper

Mapper processes input data and converts it into key-value pairs.

What is Reducer

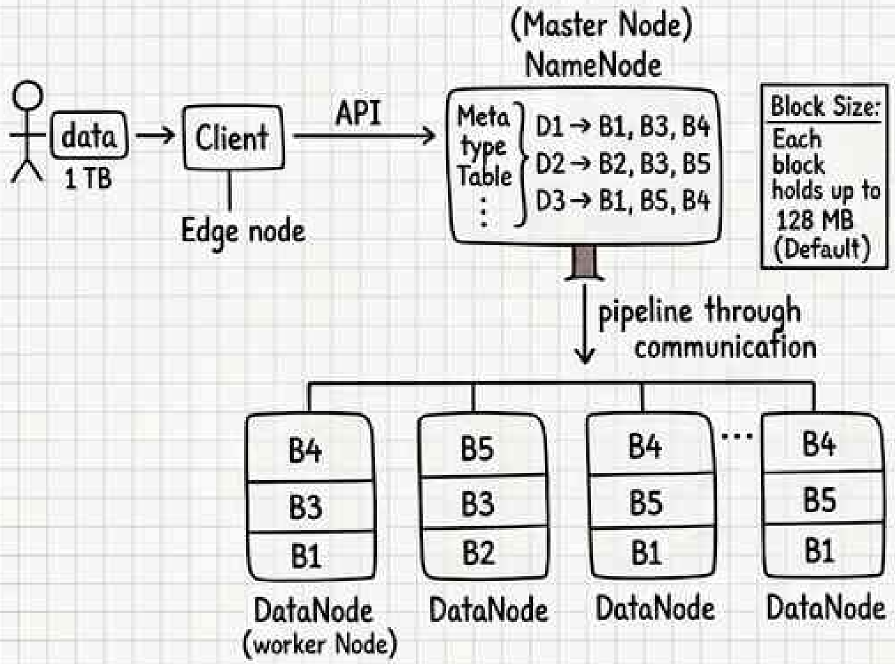
Reducer combines intermediate data and produces final output

What is parallel processing ?

Parallel processing means multiple systems process data simultaneously.

Why MapReduce is slow ?

Bcz, it repeatedly reads and writes data to disk.



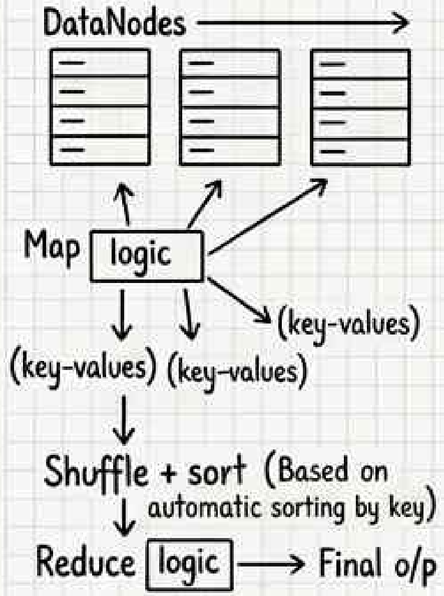
Decides on metadata of file, how the data can be distributed and where and all it can be stored.
Once it plans, through pipeline gets the data from client and stores in DataNode. Minimum it replicates each block of data for 3 times for fault tolerance

Once DataNode is done with storing the blocks it confirms to NameNode through pipeline every 3 sec. If it's heartbeat (no response) is missing, NameNode waits 10 minutes before officially marking it dead. and asks the nearby DataNode to send COPY

Map Reduce

- Map phase: logic for main purpose
- Reduce phase: logic of combining the results/data

Data will not come to logic, logic itself will go to data to save the time



Map Reduce was handling everything Resource management, container allocation, map/reduce logic
↓ (to much work)

Solution

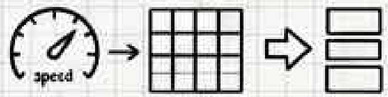
- YARN (focus on resource management)
- Map Reduce (focus on logic)

Yet Another Resource Negotiator

Components

- Application Master (Manages the complete lifecycle of a single application/job).
- Resource Manager (Allocates cluster resources to different applications).
- Node Manager (Launches and monitors resource containers on an individual machine).

Once it is set then MapReduce works



MapReduce logic goes to dataNode to perform operation & stores the temporary result in disk (permanent) and repeats for all data → Slower